

Terrorism's Echo: How Siren Alerts Impact Electoral Preferences in Israel

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Abstract

This paper relies on data from rocket alert sirens in Israel to analyse the impact of terror attacks on citizens' electoral preferences. Additionally, it investigates the effects of the siren system per se on voting patterns. By examining the evolution of the range of rockets fired from the Gaza Strip, we are able to isolate localities that remained unaffected by rocket attacks until 2014. We then employ a Differences-in-Differences approach to compare the electoral results from the 2013 and 2015 legislative elections in localities targeted versus not targeted by rockets. Our findings reveal a 1.9% to 3.0% increase in the vote share of the right-wing incumbent party Likud as a result of experiencing a siren alert between the two elections. This effect is amplified when the alarm occurs closer to the election date. Future research will aim to decompose this effect between ideological shifting and satisfaction with security provision by the state.

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1 Introduction

The Israel-Palestine conflict stands as a complex and ever-evolving issue. Since 2001, Palestinian radical organizations have launched thousands of rockets and mortars from the Gaza Strip towards Israel, with a notable surge after the extremist group Hamas assumed power in 2007 ([Ganor, 2013](#)).

Despite the effectiveness of the Iron Dome antimissile system in intercepting rocket attacks, the Israeli government implemented a comprehensive siren system in all its cities and settlements. In the face of a rocket threat, sirens are activated, and the government issues online alerts, providing vital information such as the time, date, and the names of the targeted cities. When the alarm sounds, citizens are instructed to seek shelter or move to a safe area.

Based on the evolution in the range of rockets fired from the Gaza Strip in 2014, we are able to identify the cities that were previously beyond reach and became potential targets for the first time.

Using data from official siren alarms (“Red Alerts”), we apply a Differences-in-Differences approach to these localities, categorizing them based on whether they experienced any Red Alerts and comparing their voting patterns in the 2013 and 2015 legislative elections.

Results indicate that, in localities newly within the range of rocket attacks, experiencing Red Alerts before the 2015 election caused a 1.9% to 3% increase in the vote share for the right-wing governing party Likud. The impact is greater if localities are affected closer to the election date.

The underlying causes of these effects remain uncertain: while siren warnings signal terrorist threats, they also serve a public security function by notifying citizens of these dangers. Additional research is necessary to discern the extent to which this rise is attributable to ideological shifts versus the satisfactory provision of public security.

2 Context

The origins of the Israeli-Palestinian conflict date back to before Israel’s establishment in 1948. However, the present circumstances are influenced by a series of events that have led to the emergence of extremist Palestinian factions, especially within the Gaza Strip. This region is now home to around 2 million Palestinians, who face restrictions on leaving without Israeli permission.

Originally conquered by Israel from Egypt in the 1967 Six-Day War, the Gaza Strip witnessed the arrival of thousands of Israeli Jewish settlers alongside the existing Palestinian population ([Hammel, 2020](#)).

In August 2005, the Israeli government unilaterally withdrew all its citizens and military presence from the Gaza Strip, leaving only Palestinians in the area. Subsequently, the 2006 elections resulted in the victory of the extremist group Hamas, which assumed power in 2007, intensifying efforts to destroy Israel through terror attacks. This led to Israel's blockade of Gaza in 2007, imposing severe control over the movement of Palestinians and goods, as well as over its airspace and territorial waters ([Etkes and Zimring, 2015](#)).

The persistent threat of terror attacks that followed Hamas' election, especially rocket strikes, prompted the Israeli government to heavily invest in what is now arguably one of the world's most advanced defense systems. The Iron Dome, an aerial defense system that intercepts rockets and mortars midair, was deployed in 2010, successfully intercepting around 90% of all rocket attacks since then ([Kattan, 2018](#)).

This configuration remains as the status-quo until today, with tensions rising multiple times throughout the years, marked by pikes of Palestinian terror attacks and Israeli military operations in Gaza that result in multiple casualties.

In 2014, three Israeli Jewish teenagers were abducted by Hamas members in the West Bank, which led to an Israeli military operation in order to rescue them. This operation ended with hundreds of Palestinians arrested, while the teenagers were found dead.

This led to an escalation of tensions, marked by Hamas launching a series of rocket attacks towards Israel, utilizing newly acquired rockets with extended range capabilities. These rockets had the capacity to reach Israeli localities previously untouched by attacks from Gaza, although most were stopped by the Iron Dome.¹

Furthermore, every locality in Israel had been recently equipped with an extensive siren system, that provided immediate alert to inhabitants in the event of incoming rockets targeting their vicinity (referred to hereafter as "Red Alerts"). This meant that citizens were instructed to seek shelter while the Iron Dome system attempted to intercept the rockets.

When activated, the sirens are heard throughout the entire locality. This ensures that whenever a rocket is launched towards an area, all its citizens are alerted.

Red Alerts are triggered independently of whether an interception is successful, as they are activated prior to interception. A single rocket can cause Red Alerts in several localities based on their proximity. It is also possible for a single alert to be issued when multiple rockets are fired simultaneously towards the same direction. This means that Red Alerts do not reflect exactly the amount of rockets or individual localities attacked.

Israel suffered 70 deaths in this war, primarily among its military forces. Although this figure is substantial when compared to its past conflicts, only two civilian casualties

¹"Locality" is the official term used by the Israeli government to refer to any municipal unit recognized by the Ministry of Interior. This encompasses both urban and rural municipalities as well as local councils.

were caused by rocket attacks. The Red Alert warnings are credited as a major factor in keeping the casualties count low ([Kurz and Brom, 2014](#)).

In retaliation to Hamas' attacks, the Israeli military executed airstrikes on the Gaza Strip and initiated a ground operation. The conflict culminated with over 2000 Palestinians killed in Gaza, and tens of thousands injured ([Kurz and Brom, 2014](#)).

In this context, the right-wing Israeli Prime Minister at the time, Benjamin Netanyahu, saw a significant surge in popularity during the war, with his approval ratings jumping from below 50 percent to almost 80 percent ([Feinstein, 2018](#)). He and his party (Likud) remain in office until today.

We argue that this reality, where sirens are activated and prompt individuals to seek shelter from rocket attacks, significantly influences the electoral preferences of those affected by it.

3 Related Literature

This research primarily contributes to the fields of terrorism and electoral accountability. It also indirectly enhances other areas of economic literature, including the impact of violence exposure on electoral accountability and the influence of electoral accountability on the provision of public goods.

[Gould and Klor \(2010\)](#) use geographic variation in terror attacks in Israel to argue that terrorism shifts the parties' political spectrum, and is, therefore, an effective political tool. [Elster \(2019\)](#) uses claims for rocket-related property damages in Israel to show that those affected by Gaza's rockets tend to favor right-wing parties.

[Berrebi and Klor \(2008\)](#) apply the variation of terror attacks across time and space to argue that the effect of terror attacks is heterogeneous depending on its timing and the historical political leanings of the locality. This last aspect is also examined in this paper, by examining treated observations based on the timing of the attacks, as well distance relative to Gaza and frequency of attacks.

International cases, such as [Bali \(2007\)](#) (Spain) and [Kibris \(2011\)](#) (Turkey), also support the argument that terrorist attacks tend to increase support for right-wing parties.

Regarding a broader exposure of voters to violence, [Hadzic et al. \(2020\)](#) uses Bosnia's experience to show that ethnic groups are more inclined to vote for candidates from their own ethnicity and tend to exhibit greater mistrust toward candidates from other ethnicities after being affected by war.

This paper is greatly inspired by [Getmansky and Zeitzoff \(2014\)](#), who show that the advancement of rocket technology made Israeli citizens within the rockets' range more inclined to vote for right-wing parties compared to those outside of the rockets' range.

This suggests that experiencing direct rocket attacks is not necessary for being influenced by them.

Similarly, in this study, we argue that while the majority of the rockets are intercepted and most attacks result in no casualties, an indirect consequence of terror attacks - the Red Alerts and the subsequent need to seek shelter - significantly impacts citizens' behaviour and their electoral preferences.

The distinction between terror attacks and Red Alerts holds significant importance in our research focus. While existing literature predominantly examines the influence of terrorist threats and actual attacks on electoral preferences, our study centers on the activation of Red Alerts—a state-operated mechanism designed to inform and safeguard citizens against potential terror attacks.

Barro (1973) and Ferejohn (1986) argue that the electorate holds its government accountable for the provision of public goods. Considering national security as a paramount public good, and particularly pertinent for a country enduring an evergreen conflict like Israel, it is conceivable that part of the impact of Red Alerts on electoral preferences may be indicative of the government's efficacy in ensuring national security.

This suggests that the influence of Red Alerts on electoral preferences might encompass two key dimensions: Firstly, it reflects the impact over residents' inclinations towards a more hawkish government, as a consequence of the actual terror attacks. Secondly, it encompasses citizens' satisfaction with the effectiveness of national security mechanisms.

4 Data

4.1 Rocket Range

Since the first rocket attack launched from Gaza, in 2001, there has been a continuous improvement in the destructive power and overall technology of the rockets used, especially regarding their range.

While in 2008 the most advanced rocket in Hamas' arsenal could reach a maximum distance of 45km, in 2012 this range was extended to 75km. This development meant rockets launched from Gaza could reach Israel's major cities, such as Tel Aviv and Jerusalem, for the first time ever.

In 2014, the range increased again from 75km to 150km.² This indicates that prior to 2014, all cities located between 75km and 150km from Gaza were outside the rocket range of the Gaza Strip. Therefore, if they were targeted by rocket strikes thereafter, it

²<https://embassies.gov.il/MFA/AboutIsrael/Maps/Pages/Range-of-Fire-from-Gaza.aspx>
<http://www.oref.org.il/1096-en/Pakar.aspx>

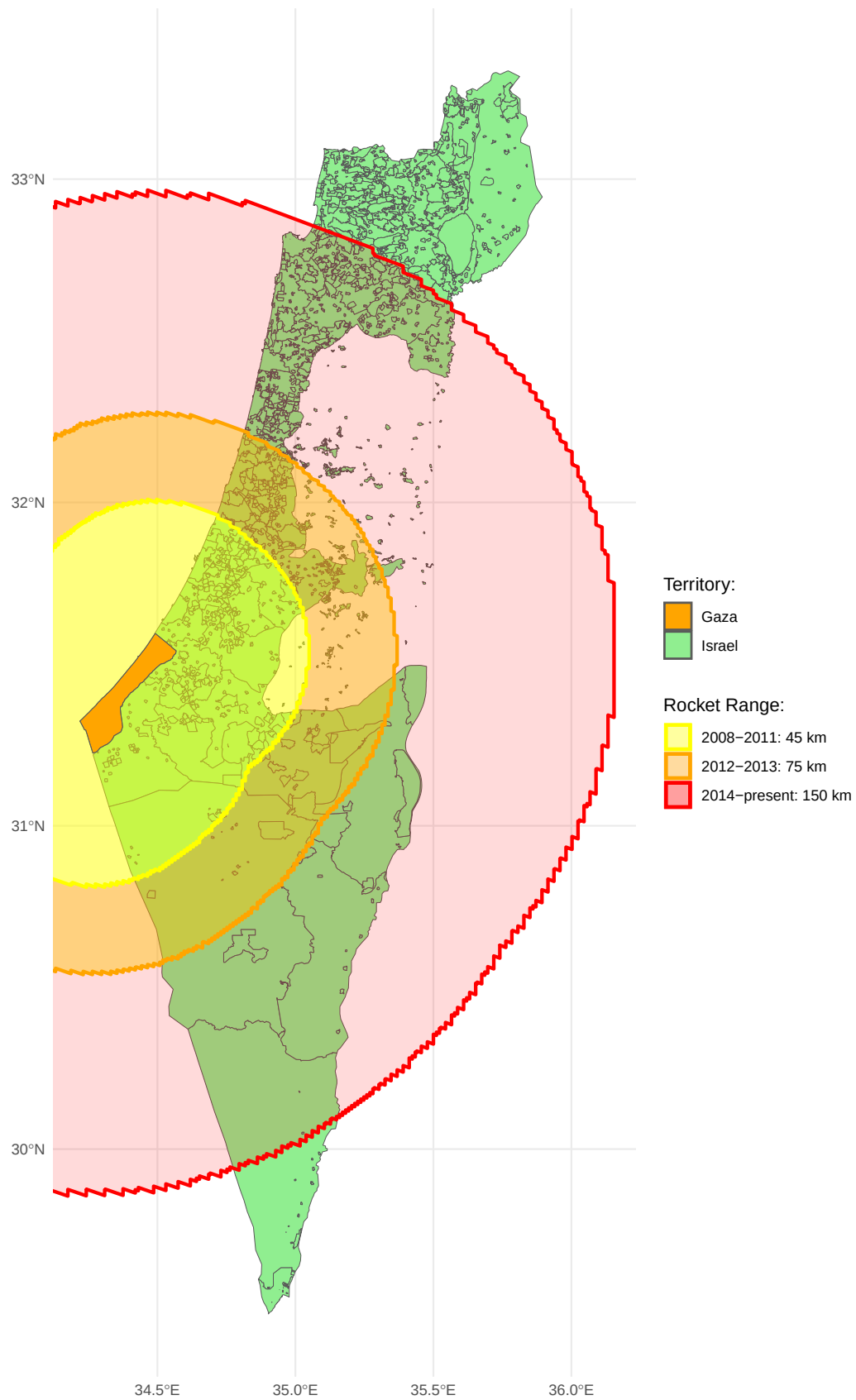
would mark the first instance of a rocket fired from Gaza towards them.

[Figure 1](#) shows the evolution of the rocket ranges fired from the Gaza Strip.

The area highlighted in red in [Figure 1](#) is the focal point of this paper, as our analysis is concentrated on the localities that entered the rockets range in 2014.

An important assumption underlying this analysis is that these localities had never been subjected to rocket attacks originating from Lebanon (to the north) or from the West Bank (to the west). This assumption is reasonable considering that the vast majority of rockets originate from the Gaza Strip, and those from Lebanon typically target localities in the far north of the country, beyond the range of Gaza. Additionally, terrorist attacks from the West Bank primarily target settlements located within the West Bank itself and typically do not involve rocket attacks. Additionally, [Getmansky and Zeitzoff \(2014\)](#) demonstrate that the evolution of the rockets' range created exogenous variation in the threat posed by terrorism.

Figure 1: Rocket Ranges from the Gaza Strip: 2008 to present



4.2 Sirens

The data on Red Alerts due to terrorist attacks was obtained from the Israeli Home Front Command website and filtered to only consider rocket alerts. Each time a rocket attack occurs, an online alert is issued simultaneously with the activation of the sirens in the targeted city and nearby localities. These alerts, containing the date, hour, and targeted cities, have been systematically collected through web scraping from the first available date, on July 2014, the beginning of the 2014 war, onwards.

It is important to note that the collected data only indicates when a rocket was fired and its target. Information on whether the rocket was intercepted is not available. Additionally, if a rocket is aimed at an uninhabited area, neither the Iron Dome system nor the Red Alert are activated, and no governmental alert is issued. Consequently, such events are not included in the database.

[Figure 2](#) maps all Red Alerts issued in Israel between the first occurrence, in July 2014, and the 2015 election. Additionally, it features two orange arcs representing distances of 75km and 150km from the Gaza Strip, delimiting the area that was in Hamas' rocket range for the first time in 2014.

As illustrated in [Figure 2](#), the majority of rocket attacks are focused on the localities close to the Gaza Strip. This pattern is logical, since targeting nearby cities is cost-effective, because rockets with shorter ranges are less expensive, and require less fuel to reach their targets. Additionally, the short distance means a shorter interval between the Red Alert and the rocket's impact, giving residents less time to find shelter.

This means that, although the localities within 75km and 150km became possible rocket targets from Gaza for the first time in 2014, the majority of the rocket attacks kept concentrated on localities near the Gaza Strip.

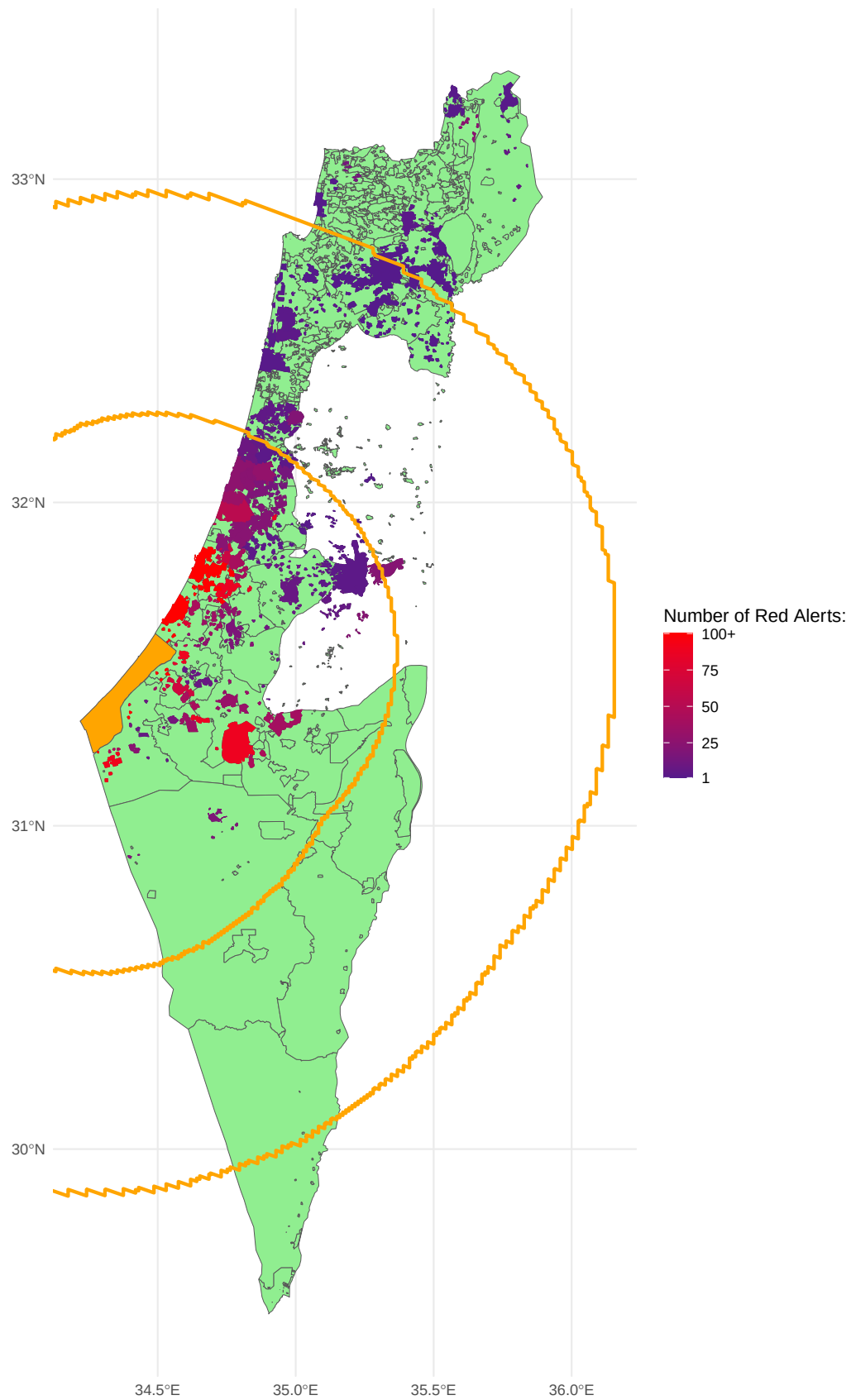
Furthermore, it is noteworthy that almost no localities south of the Gaza Strip were targeted. This can be attributed to the fact that the southern half of Israel is predominantly a desert, with fewer inhabitants.

The coordinates and polygons of all affected localities were obtained from Israel's Central Bureau of Statistics. This allows us to calculate distances between all localities and the Gaza Strip, which is crucial to explore the interaction between the attacks and distance from Gaza.

All locality-level demographic variables were also obtained from the Central Bureau of Statistics. They include demographic density, population size, main religion and age distribution of the population.

Voting results for each locality were sourced from the Israeli Central Elections Committee.

Figure 2: Red Alerts in Israel: 2014 to 2015's Election



5 Empirical Strategy

By using the evolution of the range of the rockets fired from Gaza, we can ascertain that all localities more than 75km away from Gaza had never experienced a rocket attack before July 2014, as the rockets could not reach them.

Therefore, when there was a new series of rocket attacks and a new range expansion - from 75km to 150km -, localities within these ranges became potential targets for the first time. These are the localities we consider as our observations.

Consequently, we designate the treatment group as all localities that experienced Red Alerts (which means that at least one rocket was fired in their direction) between the 2013 and the 2015 legislative elections. The initial control group consists of all Israeli localities between 75km and 150km from Gaza that had no Red Alerts between the 2013 and the 2015 legislative elections.

Figure 3 maps those groups. It should be noted that the areas in grey ranging from 75km to 150km from Gaza are non-jurisdictional zones, meaning they do not contain electoral districts.

Using a *Differences-in-Differences* approach, we compare how treated and untreated localities voted in the 2013 (before entering the rocket range) and in the 2015 (after entering the rocket range) legislative elections.³ Likud is a traditional right-wing party in Israel, and was uninterruptedly in power from 2009 to 2021, and from 2023 to today, under its leader, Benjamin Netanyahu. Due to its longstanding presence in power, we focus on the percentage of votes received by Likud.

The *Differences-in-Differences* model is estimated using the following equation:

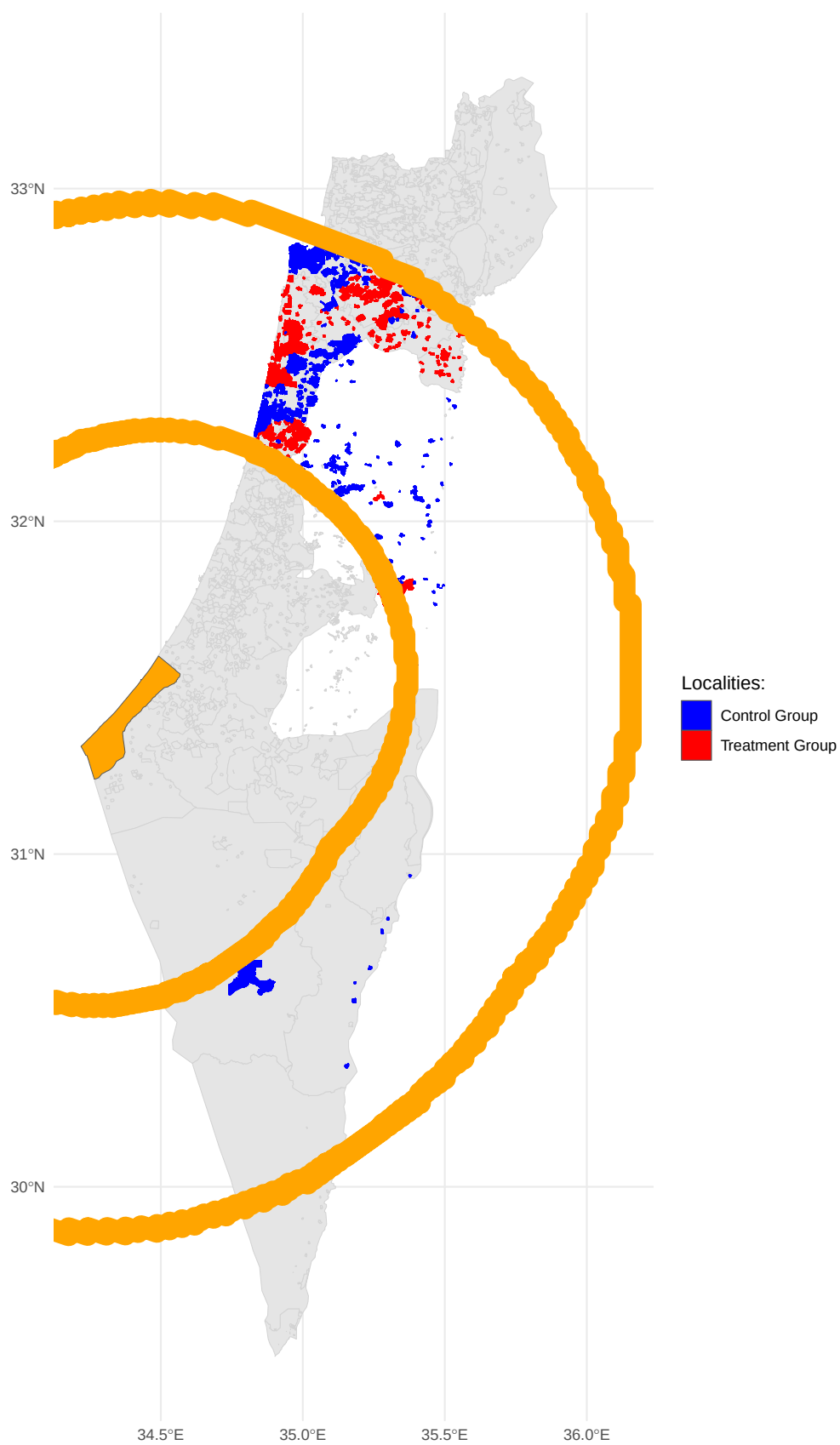
$$Likud_Percentage_{i,t} = Red_Alert_i + Post_t + Red_Alert_i \cdot Post_t + X_{i,t} \quad (1)$$

Here, $Likud_Percentage_{i,t}$ represents the Likud vote percentage in locality i during election t (2013 or 2015); Red_Alert_i is a dummy variable indicating whether there were any Red Alerts in locality i between the 2013 and the 2015 elections; $Post_t$ is a dummy variable that equals 0 for the 2013 election (pre-treatment period) and 1 for the 2015 election (post-treatment period); $X_{i,t}$ is a vector containing control variables for locality i during election t . The adopted control variables are demographic density, population size, distance from the Gaza Strip, main religion, and percentage of the locality's population above 65 years old.

$Red_Alert_i \cdot Post_t$ represents the interaction effect of how much experiencing a Red Alert impacts on the Likud share of votes in the 2015 election for locality i . It measures the difference between Likud's vote share between the 2013 and 2015 elections and

³Israel is a parliamentary state. Therefore, the Prime Minister is the head of state, and is indirectly decided as a result of the legislative election.

Figure 3: Localities between 75km and 150km: 2014 to 2015's Election



between the treated and untreated localities. This is the Differences-in-Differences estimator, and precisely the effect we want to measure.

However, it is crucial to understand that, by applying this model, we are measuring the impact of experiencing Red Alerts over Likud's vote share. Red Alerts are not merely a proxy for rocket attacks; since they also serve as a mechanism to insure civilian safety.

The root causes of Red Alert's effects on Likud's vote share are still not clear. Exposure to Red Alerts may influence residents' political ideology, potentially nudging their views towards more right-wing and militant stances. Conversely, given that Red Alerts fulfill a role in public security by alerting citizens of rocket threats, the observed effect might also encompass a general contentment with public safety measures.

6 Descriptive Statistics

We consider treatment as having at least one Red Alert between the 2013 and the 2015 elections (since the localities we are considering in our analysis became potential targets in July 2014). Our initial control group is composed of the localities that had no Red Alerts during this period.

Figure 4 presents Likud's average share of votes since the 2006 election, comparing how localities that experienced Red Alerts between the 2013 and the 2015 elections voted in relation to localities that had no Red Alerts within this period.

It is clear, from Figure 4, that before 2014, both the treatment and control group had an almost identical voting pattern. In the 2015 election, though, localities that had experienced Red Alerts between 2014 and the 2015 election voted on average, 2.2% more on Likud when compared to those that experienced no Red Alerts during this period.

Although no observation from this control group was treated before the 2015 election, many of those observations were later treated. Therefore, we could consider an alternative control group, composed only of localities that had never experienced Red Alerts from the 2013 election until the last legislative election in 2022. Figure 5 presents this alternative specification.

Both control group specifications are feasible, as we observe pre-treatment parallel trends in each case and, aside from the shifts in the initial post-treatment period (2015) that cause divergence between the treatment and control groups, the post-treatment trends remain parallel.

The comparison illustrated in Figure 5 reveals a more pronounced treatment effect, when compared to Figure 4. This is evidenced by the control group's decline in Likud's vote share from 2013 to 2015, contrasted with the treatment group's average increase

Figure 4: Likud's Average Share of Votes: 2006 to 2022

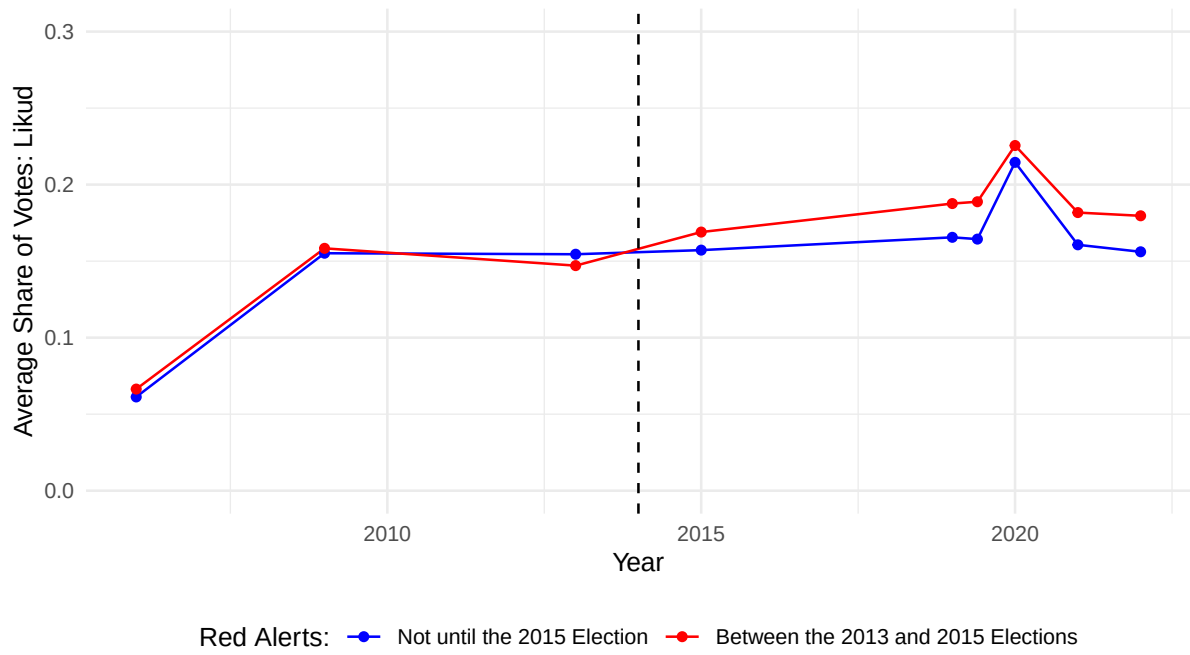
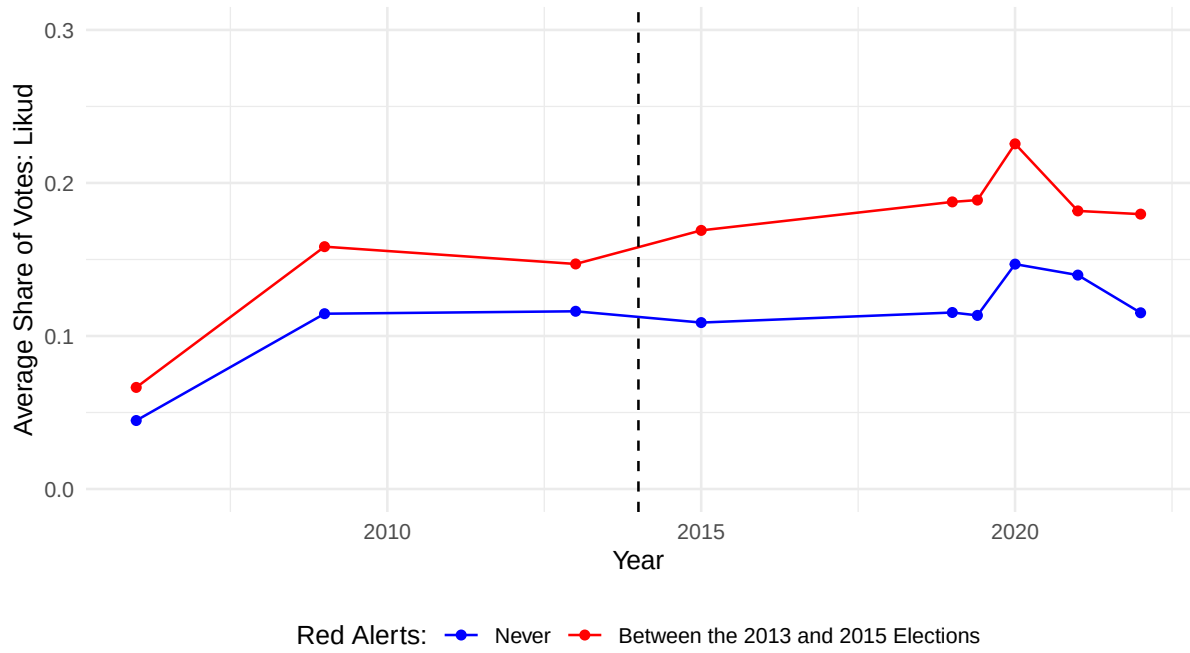


Figure 5: Likud's Average Share of Votes: 2006 to 2022 - Alternative Specification



in votes for Likud.

[Table 1](#) presents the descriptive statistics for the localities treated before the 2015 election and for both specifications of control groups (Not Treated Until the 2015 election and Never Treated) for the 2013 and 2015 elections.

Table 1: Descriptive Statistics by Groups of Interest for 2013 and 2015

Statistic	Treated			NTU 2015			Never Treated		
	2013	2015	Diff	2013	2015	Diff	2013	2015	Diff
Likud's Vote Share	14.70 (12.27)	16.90 (16.06)	2.2	15.45 (13.14)	15.72 (13.61)	0.27	11.61 (11.13)	10.88 (11.39)	-0.73
Demographic Density	1395 (1200)	1451 (1239)	56	1620 (1486)	1722 (1536)	92	2589 (2024)	2707 (2119)	118
Population Size	4516 (11658)	4667 (12003)	151	6043 (25516)	6549 (26595)	504	7665 (12884)	7913 (13320)	248
Distance to Gaza		120.27 (23.60)			107.31 (21.47)			121.14 (25.07)	
Observations		178			195			28	

NTU 2015 stands for Not Treated until the 2015 Election.

Standard Errors are reported in parenthesis.

Demographic Densities are reported in population per km².

Distances to Gaza are reported in km.

[Table 1](#) reports that, while there was on average a 2.2% rise in the Likud share of votes for the treated localities, Likud's vote share at localities that had not experienced any Red Alerts until the 2015 election rose only 0.27%, and decreased by 0.73% when considering only the localities that have never experienced Red Alerts.

On the other hand, the overall profile of the treated localities is different than expected: their demographic density is smaller than both control groups', as well as their population size. Additionally, they are on average further away from Gaza than the localities not treated until the 2015 election.

These statistics are staggering, since one would expect that rocket attacks would mainly target largely populated and high density cities, as close to Gaza as possible, since the time between the Red Alert and the rocket arrival would be smaller.

It is important to remark, however, that the statistics reported in [Table 1](#) present high standard deviations, which indicates that there is a broad spectrum of localities contained at each group. This means that treated localities are not that different from localities from the control groups, given the large intragroup variances. Furthermore, there are only 28 localities that had never experienced a Red Alert until the last legislative election (2022) composing the Never Treated control group.

7 Results

By regressing Likud’s vote percentage from each locality in the 2013 and 2015 elections, as shown in [Model 1](#), we are able to test whether experiencing at least one Red Alert prior to the 2015 election influences the vote share that the Likud party receives in the impacted locality for that election.

To ensure robustness, we calculate the Differences-in-Differences estimator using various specifications. These range from the canonical model, that includes only the dummy for treatment, the period and their interaction ([Helter, 2022](#)) to more complex specifications that include locality fixed effects and the control variables described in [section 4](#). The results for all specifications are presented in [Table 2](#).

Table 2: Differences-in-Differences Estimates: Red Alert Impact on Likud’s Vote Share, 2013 and 2015 Elections

	(1)	(2)	(3)	(4)	(5)	(6)
Red Alert · Post	0.019** (0.006)	0.020* (0.010)	0.018* (0.009)	0.029*** (0.007)	0.030** (0.011)	0.028** (0.009)
Red Alert	−0.007 (0.013)	0.078*** (0.005)	0.093** (0.031)	0.031 (0.023)	−0.150*** (0.005)	−0.756 (0.473)
Post	0.003 (0.004)	0.006 (0.007)	0.005 (0.008)	−0.007 (0.005)	−0.004 (0.008)	−0.003 (0.009)
Intercept	0.155*** (0.009)	0.087*** (0.003)	−0.569 (0.503)	0.116*** (0.021)	0.297*** (0.004)	0.757+ (0.458)
Observations	374	332	373	206	179	206
Control Variables	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
Control Group	NTU 2015	NTU 2015	NTU 2015	Never treated	Never treated	Never treated
Locality Fixed Effects	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
Includes Arab Cities	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE
Clustered Errors	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

Notes: NTU 2015 stands for Not Treated until the 2015 Election, and is composed of localities that had not experienced Red Alerts until the 2015 Election. The Never Treated control group is composed of localities that experienced no Red Alerts until the last legislative election, in 2022. Columns (1)-(3) use NTU 2015 as the control group, while columns (4)-(6) use Never Treated as the control group. Columns (2)-(3) and (5)-(6) include locality fixed effects. Columns (2) and (4) exclude all Arab localities. Columns (3) and (6) include demographic density, population size, distance to the Gaza Strip and percentage of population above 65 years old as control variables. The coefficients of the control variables and locality fixed effects are omitted from the table in order to save space. Standard Errors are reported in parenthesis and clustered by locality-level for all columns.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

From [Table 2](#), we can see that the Differences-in-Differences estimator (*Red_Alert · Post*) ranges from 1.9% to 3.0%. This indicates that a locality experiencing a Red Alert between the 2013 and the 2015 elections would, on average, increase their vote share

on Likud from 1.9% to 3.0% compared to if they had not experienced a Red Alert during that period (their counterfactual).

All specifications are robust to at least a 5% significance level, and present Differences-in-Differences estimators of similar magnitudes, which reinforces the results' robustness. The reported values are also similar to what is reported in the related literature, such as [Getmansky and Zeitzoff \(2014\)](#) (2%-6%), [Elster \(2019\)](#) (4%) and [Berrebi and Klor \(2008\)](#) (1.35%).

This outcome is arguably more pertinent and accurate, as Red Alerts occur far more frequently than actual rocket strikes and victims.

The main differential in our results is that they do not simply report the impact of rocket attacks on votes, but the effect of the Red Alerts (i.e. the activation of sirens) over votes.

The estimated impact may be attributed to a combination of elements: the electorate's inclination towards more bellicose policies; and appreciation for state-provided protection, since most of the rocket attacks were intercepted ([Kattan, 2018](#)) and the siren system helped minimize the number of injuries and casualties ([Kurz and Brom, 2014](#)). The results showcased encapsulate the sum of these effects.

Another interest result concerns the *Post* variable, which indicates whether the year is 2015 or not. Therefore, on average, the effect of going from the 2013 election to the 2015 election is not statistically different than 0. This indicates the possibility that localities experiencing Red Alerts have no significant spillover effects over untreated localities from any of the control groups.⁴

8 Timing of Red Alerts

Variations in treatment across localities suggest a potential departure from the strict adherence to the stable-unit-treatment-value-assumption (SUTVA) ([Imbens and Rubin, 2015](#)). Put simply, our definition of treatment may be overly broad. It is possible that what we call treatment actually encompasses a spectrum of similar yet not identical interventions, potentially deviating from the assumption of uniform treatment effects.

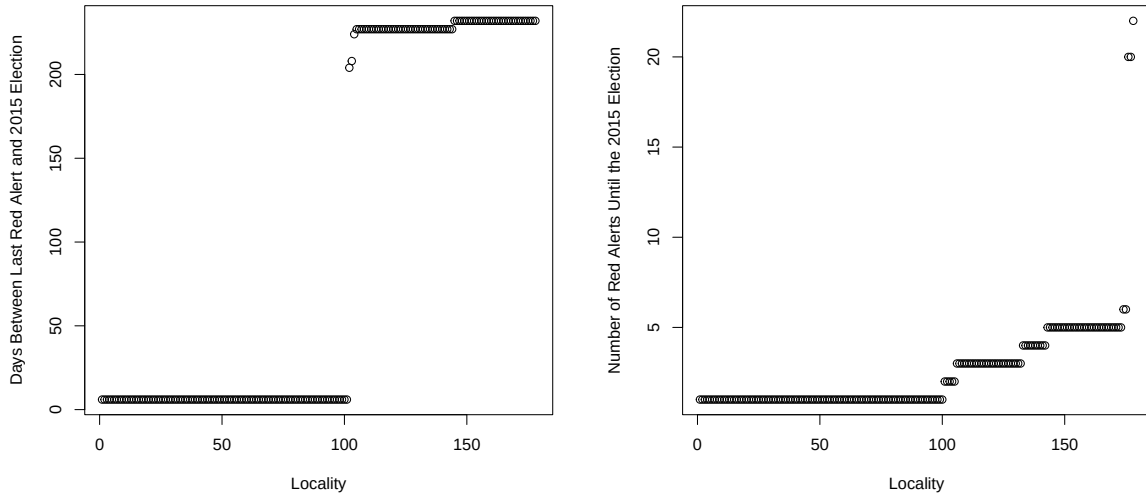
Does a Red Alert issued 7 months prior to the 2015 election have the same effect as a Red Alert issued only 6 days before the election? In order to address this subject, we need to check for potential heterogeneity on the treatment effect based on the timing of the Red Alerts. One possible approach is to divide the treatment group into subgroups

⁴For this to be true, however, we would need to assume that there are no uncontrolled factors that influence Likud's vote share, which is a considerably strong assumption, especially since, as stated in [section 5](#), there are SUTVA issues in this approach.

with similar characteristics and compare the results.⁵

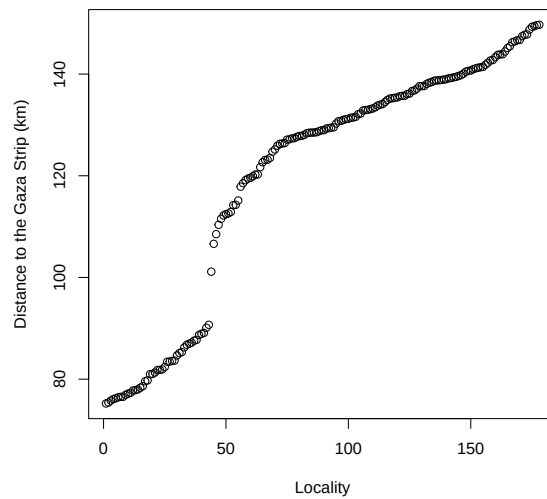
Figure 6 exposes the differences in characteristics among treated localities.

Figure 6: Visual Representation of Heterogeneity Between Treated Units



(a) Days Before 2015 Election

(b) Number of Red Alerts



(c) Distance to Gaza

⁵Alternatively, these factors could be incorporated into the regression as moderators, i.e. variables that influence the strength/degree and even direction of the relationship between the independent and dependent variables (Stone and Hollenbeck, 1984).

As it is clear in [Figure 6a](#), there is a big discontinuity regarding the number of days before the 2015 election when treated localities received their last Red Alert. Notably, 101 localities experienced their last Red Alert only 6 days prior to the election, whereas the remaining 77 localities had their last Red Alert between 204 and 232 days before the election.

[Figure 7](#) maps these red alerts based on the amount of days before the 2015 election.

Therefore, by dividing the original treatment group into these two clusters, we can observe how the proximity of the Red Alert to the election date influences the electoral preferences of the localities. [Figure 8](#) plots the evolution of Likud's average vote share based on the temporal distance between the last Red Alert issued and the 2015 Election.

As depicted in [Figure 8](#), the timing before the election seems to be determinant on voting preferences.⁶

During the pre-treatment period, both treatment groups exhibited nearly identical voting patterns and reported an average rise in Likud's vote share from the 2013 to the 2015 election. However, the magnitude of the increase in vote share for localities that experienced a Red Alert just 6 days before the election was substantially greater than that of the group that experienced it 204-232 days before the election.

To estimate the impact of experiencing a Red Alert either 6 or 204-232 days before the 2015 election, we must modify [Model 1](#). Consequently, the revised model should be:

$$Likud_Percentage_{i,t} = Last_Red_Alert_i + Post_t + Last_Red_Alert_i \cdot Post_t + X_{i,t} \quad (2)$$

In [Model 2](#), the only difference to [Model 1](#) is that, instead of discriminating between treated and untreated localities, we use $Last_Red_Alert_i$. This is a categorical indicator that indicates whether locality i was treated 6 days before, 204-232 days before and not treated before the 2015 election.

⁶The Control Group used in [Figure 8](#) is composed of "Never Treated" localities, i.e. localities that have never experienced a Red Alert (until the last legislative election, in 2022). A similar graph with the Control Group as localities that had not experienced a Red Alert until the 2015 election can be found in [Appendix A](#)

Figure 7: Red Alerts in Israel based on Distance to 2015's Election: 75-150km from the Gaza Strip

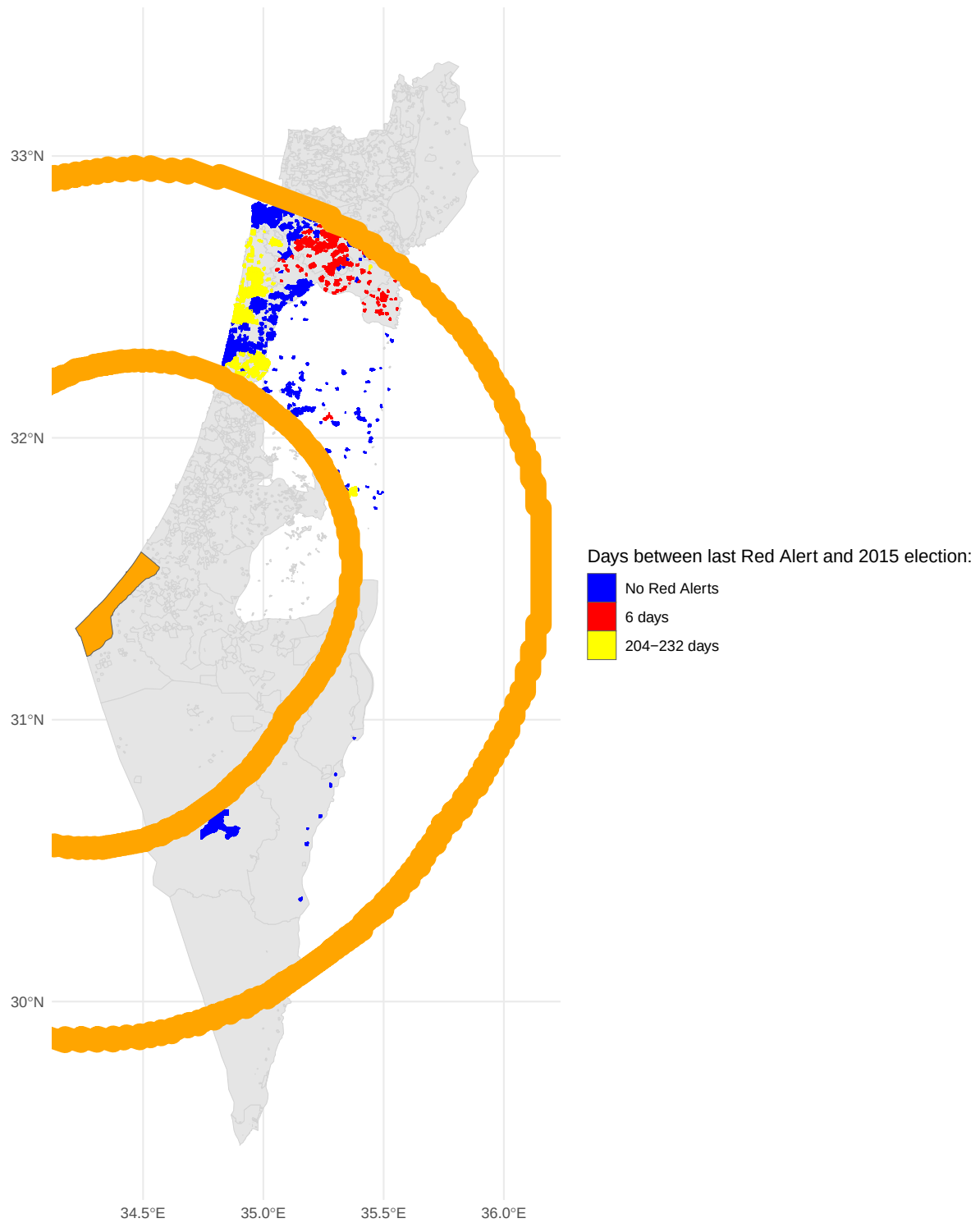


Figure 8: Likud's Average Share of Votes: 2006 to 2022 - Days Between Last Red Alert and 2015 Election

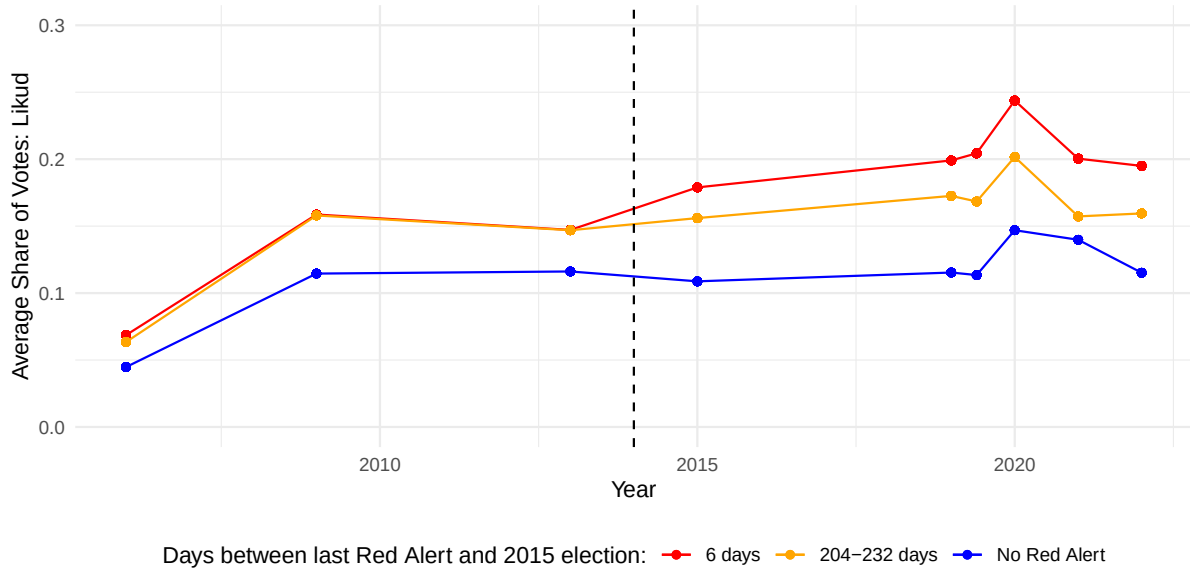


Table 3 displays the outcomes of the same regressions from Table 2, but distinguishes the treatment group into localities that experienced Red Alerts six days prior to the 2015 election and those that were exposed to Red Alerts between 204 and 232 days before the 2015 election.

Discriminating with respect to the timing of the Red Alert reveals that its impact on Likud's vote share significantly fluctuates based on the days separating the alert from the election. As indicated in Table 3, experiencing a Red Alert six days before the election boosts Likud's vote share by an average of 2.9% to 4.1%. This effect is more pronounced than the one presented in Table 2, where the temporal proximity of the Red Alert was not taken into account.

On the other hand, a Red Alert 204-232 days before the 2015 election had no significant effect over Likud's vote share. This implies that the impact of experiencing a Red Alert on a locality's electoral preferences decreases over time. Therefore, with enough time between the Red Alert and the election, a treated unit could become identical to its untreated counterfactual.⁷

In Appendix A, we conduct a complementary analysis exploring additional sources of treatment heterogeneity. Specifically, we investigate whether the number of Red Alerts experienced by a treated locality between the 2013 and 2015 elections influences Likud's vote share. Furthermore, we examine whether the proximity of localities to

⁷The practical effects of this finding means that our observations do not follow a staggered adoption setting, where once they are treated they remain equally treated forever. There are many relevant implications of this finding, such as the impossibility to adopt staggered Differences-in-Differences in the future, such as Callaway and Sant'Anna (2021) event-study (Callaway and Sant'Anna, 2021; Helter, 2022).

Table 3: Differences-in-Differences Estimates: Red Alert Impact on Likud's Vote Share (Subdivided by Alert Timing), 2013 and 2015

	(1)	(2)	(3)	(4)	(5)	(6)
Red Alert						
6 Days Before * Post	0.029*** (0.008)	0.031** (0.012)	0.028* (0.011)	0.039*** (0.008)	0.041** (0.013)	0.037** (0.011)
Red Alert						
204+ Days Before * Post	0.006 (0.007)	0.006 (0.011)	0.006 (0.010)	0.016* (0.007)	0.015 (0.012)	0.016 (0.010)
Red Alert						
6 Days Before	-0.007 (0.016)	0.325*** (0.006)	0.202+ (0.117)	0.031 (0.025)	0.115*** (0.006)	-0.128 (0.339)
Red Alert						
204+ Days Before	-0.008 (0.015)	0.085*** (0.005)	0.098** (0.030)	0.031 (0.024)	-0.143*** (0.006)	-0.590 (0.413)
Post	0.003 (0.004)	0.006 (0.007)	0.005 (0.008)	-0.007 (0.005)	-0.004 (0.008)	-0.004 (0.009)
Intercept	0.155*** (0.009)	0.087*** (0.003)	-0.506 (0.463)	0.116*** (0.021)	0.297*** (0.004)	0.530* (0.251)
Observations	374	332	374	206	179	206
Control Variables	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
Control Group	NTU 2015	NTU 2015	NTU 2015	Never treated	Never treated	Never treated
Locality Fixed Effects	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
Includes Arab Cities	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE
Clustered Errors	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

Notes: NTU 2015 stands for Not Treated until the 2015 Election, and is composed of localities that had not experienced Red Alerts until the 2015 Election. The Never Treated control group is composed of localities that experienced no Red Alerts until the last legislative election, in 2022. Columns (1)-(3) use NTU 2015 as the control group, while columns (4)-(6) use Never Treated as the control group. Columns (2)-(3) and (5)-(6) include locality fixed effects. Columns (2) and (4) exclude all Arab localities. Columns (3) and (6) include demographic density, population size, distance to the Gaza Strip and percentage of population above 65 years old as control variables. The coefficients of the control variables and locality fixed effects are omitted from the table in order to save space. Standard Errors are reported in parenthesis and clustered by locality-level for all columns.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

the Gaza Strip acts as a moderating factor in this context. Regarding both the distance from Gaza and the number of Red Alerts, we find no evidence to suggest they affect the treatment.

9 Next Steps

Although the results presented in this paper exhibit statistical significance, robustness, and magnitudes consistent with existing literature, our analysis is somewhat constrained by its focus solely on the 2013 and 2015 elections. By adopting an event-study methodology, we could extend our investigation to encompass all subsequent elections, thereby expanding the temporal scope of our findings.

The refinement of siren precision in 2019 is particularly interesting, wherein Red Alerts were tailored to specific localities rather than encompassing nearby areas. This notable discontinuity presents a compelling opportunity for exploration if we extend our analysis to encompass the most recent elections.

Furthermore, conducting a Differences-in-Differences analysis similar to this paper's on other right-wing parties and comparing the outcomes with those of Likud could provide deeper insights into whether the electorate from attacked localities shifted their support to the right-wing in general or only to Likud.

Such an approach would enable us to disentangle the effects of Red Alerts on electoral preferences stemming from ideological positions vis-à-vis terrorism from those driven by satisfaction with the provision of public security by the incumbent party.

Furthermore, examining whether historically left-wing cities were affected differently than traditionally right-wing cities by Red Alerts could provide valuable insights, such as whether Red Alerts increase political polarization. Additionally, analyzing voter turnout would offer further avenues for investigation.

By considering these additional variables and extending our analysis to encompass a broader temporal and geographic scope, we can enhance the depth and breadth of our understanding of the complex interplay between Red Alerts, electoral preferences, and broader political dynamics in Israel.

10 Conclusion

We used the 2014 evolution of Hamas' rocket range to determine which localities became potential targets for the first time. Using locality-level data from Israel's siren alarms (Red Alerts) that were activated whenever a city was targeted, we were able to utilize a Differences-in-Differences approach to evaluate the impact of experienc-

ing a Red Alert for the first time over the incumbent party's (Likud) vote share in the affected localities.

Results showed that experiencing for the first time Red Alerts between the 2013 and 2015 elections led to an increase in Likud's vote share between 1.9% and 3.0%. When considering only Red Alerts that occurred 6 days prior to the 2015 election, the estimated impact ranges from 2.9% to 4.1%.

Future research will be able to deepen our understanding on how Red Alerts affect overall voting behaviour in Israel. Next steps include analysing the influence of Red Alerts on the vote shares of other right-wing parties. This will clarify whether the affected voters shifted towards the right-wing in general or were simply satisfied with the security policy under Likud's government.

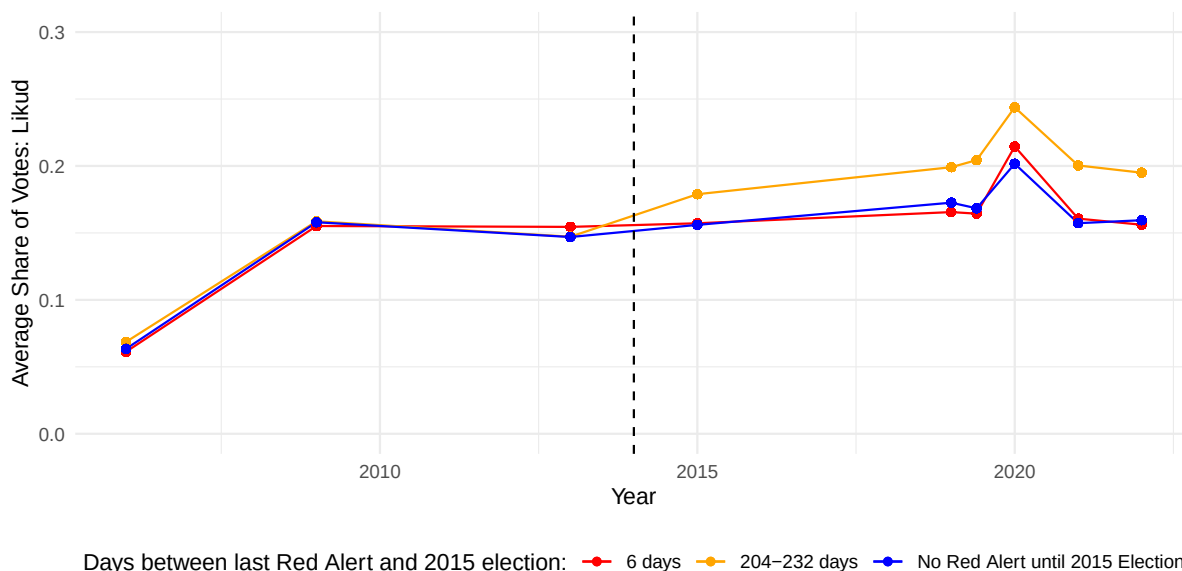
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A Appendix: Treatment Heterogeneity

Figure 9: Likud's Average Share of Votes: 2006 to 2022 - Days Between Last Red Alert and 2015 Election (Alternative Control Group)



A.1 Number of Red Alerts

As depicted in [Figure 6b](#), the majority of the treated localities recorded only 1 Red Alert prior to the 2015 election. The rest of the localities experienced up to 6 Red Alerts, with an exception of six localities that encountered 20.

Due to the small number of observations on the right side of this discontinuity, dividing the localities in groups is not viable. In this case, A better way to check for heterogeneity would be to include the number of Red Alerts as a moderator. [Model 3](#) present the adequate model.

$$Likud_Percentage_{i,t} = Number_of_Red_Alerts_i + Post_t + Number_of_Red_Alerts_i \cdot Post_t + X_{i,t} \quad (3)$$

The main difference from [Model 1](#) is $Number_of_Red_Alerts_i$, that equals how many Red Alerts locality i experienced between the 2013 and 2015 elections.

[Table 4](#) presents the results of [Model 3](#)'s model.

It is clear, from [Table 4](#), that the number of Red Alerts a locality was exposed to between the 2013 and 2015 elections does not significantly impact Likud's vote share in the 2015 election in that locality.

Table 4: Differences-in-Differences Estimates: Red Alert Impact on Likud's Vote Share (Moderated by Number of Red Alerts), 2013 and 2015

	(1)	(2)	(3)	(4)	(5)	(6)
Number of Red Alerts * Post	0.001 (0.001)	0.002 (0.002)	0.001 (0.001)	0.000 (0.001)	−0.001 (0.003)	0.000 (0.001)
Number of Red Alerts	−0.002 (0.004)	0.021*** (0.001)	0.019** (0.006)	−0.002 (0.005)	0.018*** (0.001)	0.242* (0.095)
Post	0.010** (0.004)	0.013* (0.006)	0.014* (0.007)	0.018*** (0.005)	0.024** (0.008)	0.022* (0.009)
Intercept	0.153*** (0.008)	0.083*** (0.003)	−0.961+ (0.554)	0.103*** (0.021)	0.094*** (0.004)	−0.056 (0.092)
Observations	374	332	374	206	179	206
Control Variables	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
Control Group	NTU 2015	NTU 2015	NTU 2015	Never treated	Never treated	Never treated
Locality Fixed Effects	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
Includes Arab Cities	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE
Clustered Errors	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

Notes: NTU 2015 stands for Not Treated until the 2015 Election, and is composed of localities that had not experienced Red Alerts until the 2015 Election. The Never Treated control group is composed of localities that experienced no Red Alerts until the last legislative election, in 2022. Columns (1)-(3) use NTU 2015 as the control group, while columns (4)-(6) use Never Treated as the control group. Columns (2)-(3) and (5)-(6) include locality fixed effects. Columns (2) and (4) exclude all Arab localities. Columns (3) and (6) include demographic density, population size, distance to the Gaza Strip and percentage of population above 65 years old as control variables. The coefficients of the control variables and locality fixed effects are omitted from the table in order to save space. Standard Errors are reported in parenthesis and clustered by locality-level for all columns.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A.2 Distance to Gaza

Distances between localities and the Gaza Strip are important to determine the amount of time between the Red Alert and the potential rocket arrival, i.e. the amount of time to seek shelter. However, this does not vary much for localities between 75km and 150km (approximately one minute).

In order to test whether the distance between a locality and the Gaza Strip influences the magnitude of the treatment effect, we need to include distance as a moderator (Stone and Hollenbeck, 1984).

$$\begin{aligned} Likud_Percentage_{i,t} = & Red_Alert_i + Post_t + Distance_i \cdot Red_Alert_i \cdot Post_t \\ & + Distance_i \cdot Red_Alert_i + Distance_i \cdot Post + Distance_i + X_{i,t} \end{aligned}$$

where $Distance_i$ represents the distance (in km) between locality i and the Gaza Strip. The other variables represent the same as explained in Model 1. Our interest lies in the interaction between Red_Alert_i , $Post_t$ and $Distance_i$, since it measures the impact of experiencing a Red Alert on the 2015 Likud's vote share considering how far away from Gaza is the treated locality.

Table 5 presents the results from this model.

The moderation of the distance between the locality and the Gaza strip over the treatment does not seem to impact significantly Likud's vote share.

This was expected, since the time to seek shelter is similar for all localities in the 75km-150km range.

Table 5: Differences-in-Differences Estimates: Red Alert Impact on Likud's Vote Share (Moderated by Distance Between Locality and the Gaza Strip), 2013 and 2015

	(1)	(2)	(3)	(4)	(5)	(6)
Red Alerts * Post						
* Distance	0.001** (0.000)	0.001+ (0.000)	0.001* (0.000)	0.001** (0.000)	0.001* (0.000)	0.001 (0.000)
Red Alerts * Post	-0.050+ (0.026)	-0.044 (0.040)	-0.051 (0.037)	-0.055+ (0.031)	-0.063 (0.047)	-0.042 (0.046)
Red Alerts * Distance	0.001* (0.001)			0.000 (0.001)	0.000 (0.001)	
Post * Distance	-0.001*** (0.000)	0.000+ (0.000)	-0.001* (0.000)	-0.001** (0.000)	-0.001* (0.000)	-0.001+ (0.000)
Red Alerts	-0.137* (0.063)	0.082*** (0.005)	0.091** (0.029)	0.029 (0.107)	-0.662 (0.414)	-0.756+ (0.451)
Post	0.060*** (0.018)	0.049+ (0.027)	0.065* (0.027)	0.065** (0.024)	0.068+ (0.037)	0.059 (0.038)
Distance	-0.001*** (0.000)	0.000** (0.000)	0.006 (0.004)	0.000 (0.001)	0.002+ (0.001)	0.001 (0.000)
Intercept	0.303*** (0.046)	0.127*** (0.013)	-0.515 (0.387)	0.137 (0.098)	0.352 (0.344)	0.864+ (0.438)
Observations	374	332	374	206	179	206
Control Variables	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE
Control Group	NTU 2015	NTU 2015	NTU 2015	Never treated	Never treated	Never treated
Locality Fixed Effects	FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
Includes Arab Cities	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE
Clustered Errors	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

Notes: NTU 2015 stands for Not Treated until the 2015 Election, and is composed of localities that had not experienced Red Alerts until the 2015 Election. The Never Treated control group is composed of localities that experienced no Red Alerts until the last legislative election, in 2022. Columns (1)-(3) use NTU 2015 as the control group, while columns (4)-(6) use Never Treated as the control group. Columns (2)-(3) and (5)-(6) include locality fixed effects. Columns (2) and (4) exclude all Arab localities. Columns (3) and (6) include demographic density, population size, distance to the Gaza Strip and percentage of population above 65 years old as control variables. The coefficients of the control variables and locality fixed effects are omitted from the table in order to save space. Standard Errors are reported in parenthesis and clustered by locality-level for all columns.

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$